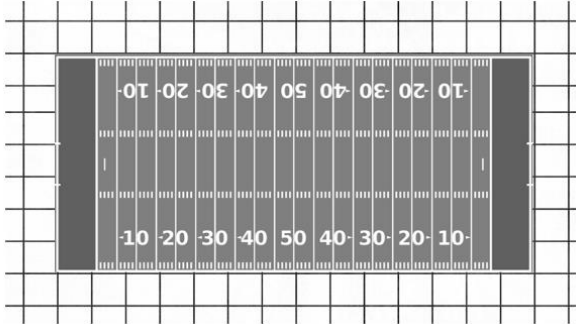


Grade 7
Unit 5
Lesson 8 Practice

Name Answer key
Date _____
Period _____

A scale drawing of the school's football field was drawn with a scale of 2 cm to 52 feet. Each gridline equals 2 cm. Use this drawing to answer questions 1-5.



1. Find the dimensions of the scale drawing (width and length).

14 cm x 28 cm

2. Find the dimensions of the actual football field (width and length).

(7 x 52) (14 x 52)
364 ft x 728 ft

3. Compare the perimeter of the scale drawing to the perimeter of the actual football field.

Drawing perimeter: $2(14) + 2(28) = 84$ cm
Real field perimeter: $2(364) + 2(728) = 2,184$ ft

4. Determine a ratio that compares the width of the scale drawing to the width of the actual football field.

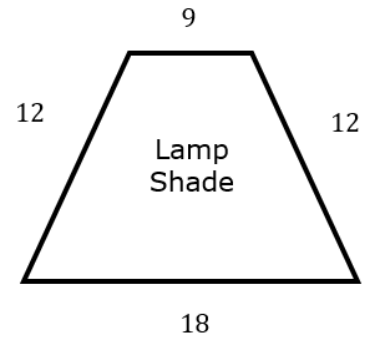
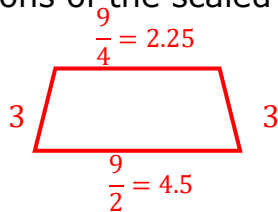
$$\frac{2 \text{ cm}}{52 \text{ ft}} = \frac{1 \text{ cm}}{26 \text{ ft}}$$

5. Determine a ratio that compares the perimeter of the scale drawing to the perimeter of the actual football field

$$\frac{84 \text{ cm}}{2,184 \text{ ft}}$$

Terrica wants to buy a lamp shade to match her bedroom. The measurements of the lamp (in inches) are shown below. Terrica wants to create a scaled drawing of the lamp using a scale factor of $\frac{1}{4}$. Use the figure to answer questions 6-7.

6. Find the dimensions of the scaled drawing of the lamp.

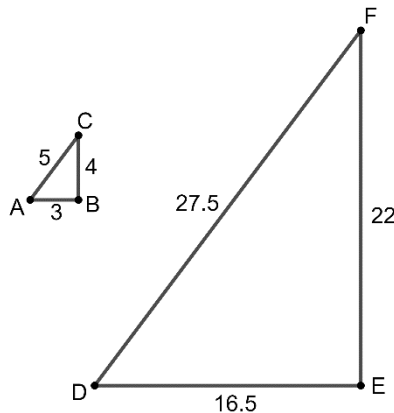


7. How does the perimeter of the scaled drawing compare to the perimeter of the actual lamp?

Drawing: $3 + 3 + 2.25 + 4.5 = 12.75$ in
 $\frac{12.75}{51} = \frac{1}{4}$ scale factor

Real: $12 + 12 + 9 + 18 = 51$ in

The dimensions of Triangle ABC and Triangle DEF are shown in the figure below (in centimeters). Use the figure to complete questions 8-10.



8. Find the ratios of 2 corresponding sides of Triangle ABC and Triangle DEF .

$\frac{3}{16.5}$; $\frac{4}{22}$; $\frac{5}{27.5}$

All simplify to $\frac{2}{11}$

9. Find the ratios of the perimeters of Triangle ABC and Triangle DEF .

$\Delta ABC: 3 + 4 + 5 = 12$

$\Delta DEF: 16.5 + 22 + 27.5 = 66$

$\frac{12}{66} = \frac{2}{11}$

10. Use the ratios in number #8 and #9 to determine if the triangles are examples of scale models. Explain your answer.

Yes, because the ratios are all the same.